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Availability of Data and Material

The datasets generated or analyzed during the study are available from the corresponding author on reasonable request.

Conflicts of Interest

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INTRODUCTION

Percutaneous endoscopic gastrostomy (PEG) is widely performed for patients requiring long-term enteral nutrition and is generally considered a safe procedure. However, complications such as infection, bleeding, aspiration, and visceral injury have been reported.¹ Air embolism is a rare but potentially fatal complication of gastrointestinal endoscopy. A systematic review of endoscopy-related air embolism demonstrated that most cases involve venous air embolism affecting the pulmonary circulation, whereas systemic embolization, particularly

Cerebral Air Embolism Occurring During Percutaneous Endoscopic Gastrostomy

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Air embolism is a rare but potentially fatal complication of gastrointestinal endoscopic procedures, primarily affecting the pulmonary circulation. Systemic air embolism, especially with cerebral involvement, is exceedingly uncommon. We report a case of cerebral air embolism that occurred during percutaneous endoscopic gastrostomy (PEG) using the pull method in a 65-year-old man with dysphagia due to progressive supranuclear palsy with frontotemporal dementia. Immediately after guidewire externalization, the patient experienced sudden respiratory arrest and hemodynamic collapse, necessitating cardiopulmonary resuscitation. Although aspiration was initially suspected, chest radiography showed no abnormalities. As spontaneous respiration did not recover, brain CT was performed, revealing multiple intracranial air densities consistent with cerebral air embolism. Transthoracic echocardiography showed no evidence of a right-to-left shunt. The patient received hyperbaric oxygen therapy; however, impaired consciousness persisted, likely due to hypoxic brain injury. This case suggests that excessive air insufflation and vascular injury during PEG may cause systemic air embolism even in the absence of an identifiable intracardiac shunt. Clinicians should consider this rare but life-threatening complication when unexplained cardiopulmonary collapse occurs during PEG.

Keywords Percutaneous endoscopic gastrostomy; Air embolism; Hyperbaric oxygen therapy.

to the cerebral circulation, is uncommon.² In normal physiology, air entering the venous system is filtered by the pulmonary vasculature; therefore, cerebral air embolism typically requires either right-to-left shunting or overwhelming of the pulmonary filtration mechanism.³ We report a rare case of cerebral air embolism that occurred during PEG insertion without evidence of an intracardiac shunt.

CASE REPORT

A 65-year-old man was referred for PEG placement due to

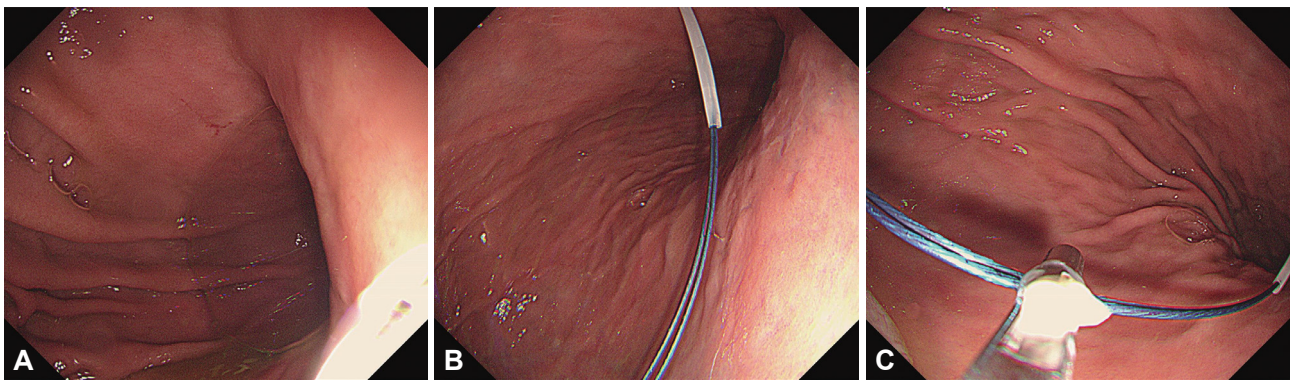


Fig. 1. Endoscopic findings during percutaneous endoscopic gastrostomy procedure. A: Gastric lumen after insufflation with identification of the puncture site. B: Guidewire entering the stomach after gastric puncture. C: Guidewire grasped and prepared for oral retrieval.

dysphagia caused by progressive supranuclear palsy with frontotemporal dementia. His medical history included benign prostatic hyperplasia and prior orthopedic surgeries for bilateral femoral fractures. The patient had previously undergone tracheostomy because of recurrent aspiration and airway protection difficulties related to advanced neurodegenerative disease. Before PEG, the patient was spontaneously breathing through the tracheostomy tube without mechanical ventilatory support and did not require continuous supplemental oxygen. Baseline laboratory findings revealed anemia (hemoglobin 9.4 g/dL) and hypoalbuminemia (albumin 2.6 g/dL), suggesting poor nutritional status. PEG was performed using the pull method without sedation. Air was used for insufflation during the procedure, and carbon dioxide was not used. To prevent interposition of adjacent organs, sufficient air insufflation was applied to achieve close apposition between the gastric and abdominal walls. The puncture site was selected using transillumination and external finger indentation, confirming close apposition between the gastric and abdominal walls. Needle puncture was then performed under direct endoscopic visualization, followed by guidewire insertion into the gastric lumen. The guidewire was grasped endoscopically and retrieved through the oral cavity for pull-method PEG placement (Fig. 1). After identification of the puncture site, gastric puncture was performed, and a guidewire was inserted into the stomach and retrieved through the oral cavity.

Immediately after externalization of the guidewire, the patient developed sudden respiratory arrest with loss of spontaneous breathing and hemodynamic collapse. Cardiopulmonary resuscitation was initiated immediately, and ventilation was performed through a pre-existing tracheostomy tube using a bag-valve device. Epinephrine was administered during resuscitation. Return of spontaneous circulation was achieved after approximately 3 minutes, and blood pressure and pulse stabilized. However, spontaneous respiration did not recover,

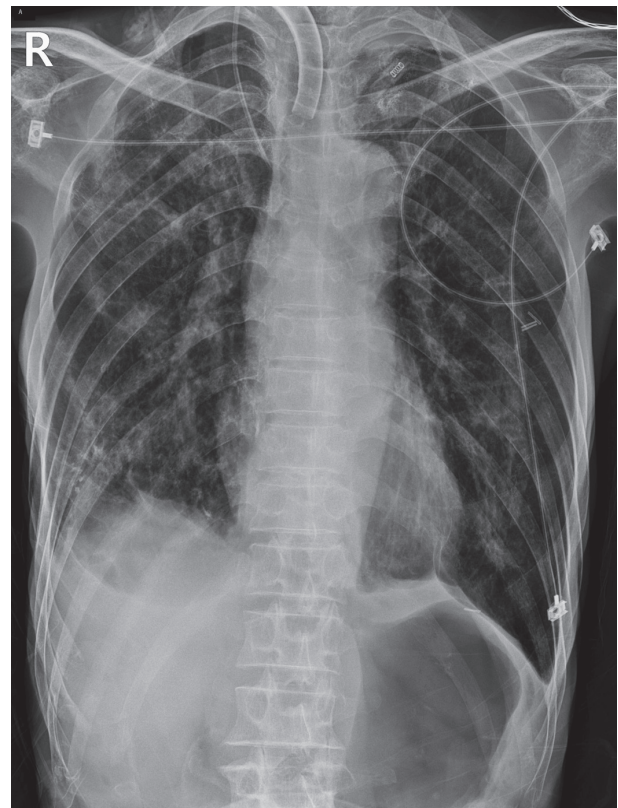


Fig. 2. Chest radiography after percutaneous endoscopic gastrostomy procedure. No newly developed infiltrates or interval change, making aspiration unlikely.

and the patient was transferred to the intensive care unit for mechanical ventilation. Although PEG tube placement had been completed using the pull method, the endoscope was not reinserted to confirm the final position of the tube because of the patient's abrupt clinical deterioration.

Given the abrupt onset of respiratory failure during PEG, aspiration was initially suspected. However, chest radiography obtained after the procedure showed no new infiltrates or sig-

nificant interval changes compared with previous imaging (Fig. 2). Because spontaneous respiration did not recover despite stable hemodynamics and unremarkable chest radiography, further neurologic evaluation was performed. Brain CT revealed multiple intracranial air densities in both cerebral hemispheres, consistent with cerebral air embolism (Fig. 3). Brain MRI did not demonstrate definite acute infarction. Based on these findings, the patient was diagnosed with cerebral air embolism. Hyperbaric oxygen therapy was initiated as standard treatment.⁴ Transthoracic echocardiography was performed to evaluate a possible right-to-left shunt; however, no intracardiac abnormality was identified. The patient's hemodynamic status gradually stabilized, and he was transferred to the general ward. However, impaired consciousness persisted, likely due to hypoxic brain injury in the context of the event and underlying neurological disease.

DISCUSSION

Air embolism during endoscopic procedures is rare but potentially fatal. It may present with cardiovascular collapse, respiratory failure, or neurologic deficits, and can be easily misdiagnosed as more common complications such as aspiration or sedation-related events.² In most cases, venous air emboli are filtered by the pulmonary circulation, resulting in pulmonary embolism rather than systemic embolization.³ However, cerebral air embolism can occur when air bypasses the pulmonary filter. The most common mechanism is intracardiac right-to-left shunting, such as a patent foramen ovale.²

In the present case, no intracardiac shunt was identified. Therefore, transpulmonary passage of air is a plausible mechanism. Experimental studies have shown that when the volume of venous air exceeds the filtering capacity of the pulmonary circulation, air can pass into the systemic arterial circulation.³ During PEG, excessive gastric insufflation may increase intra-

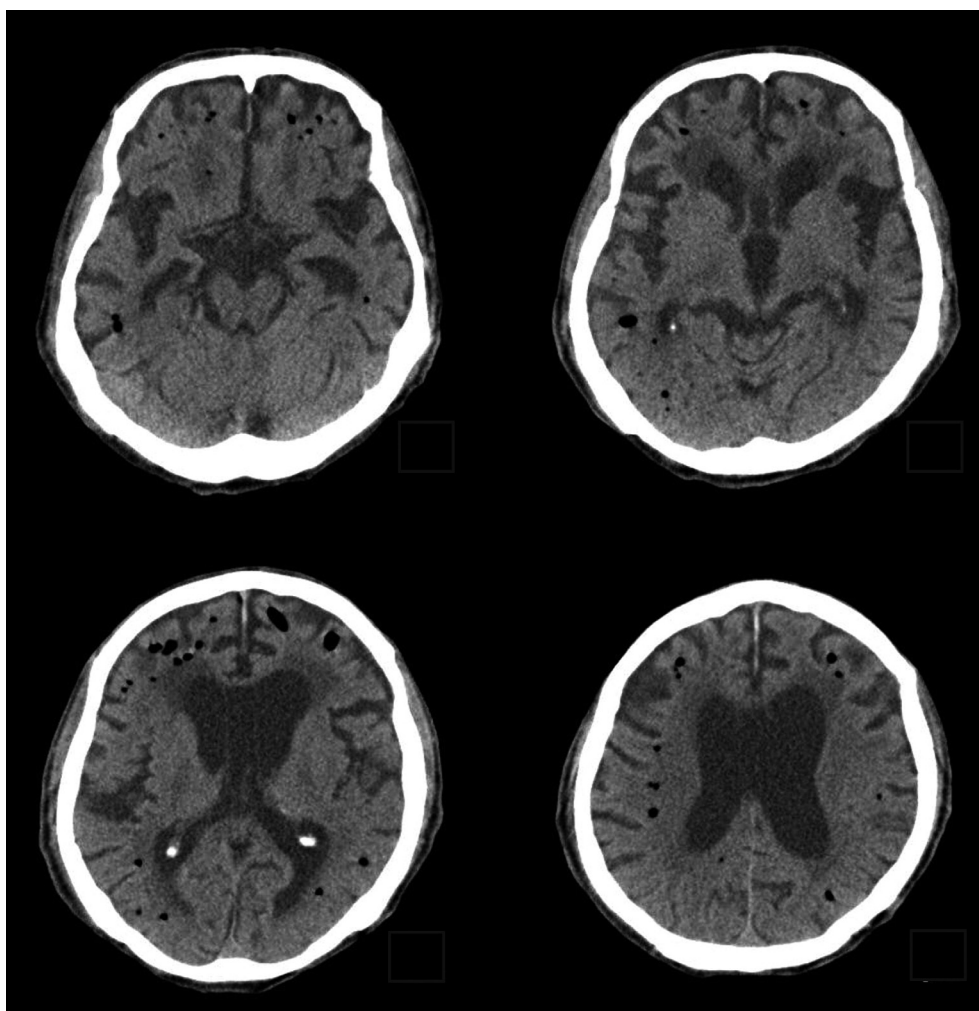


Fig. 3. Brain CT findings. Multiple intracranial air densities in both cerebral hemispheres, consistent with cerebral air embolism.

luminal pressure, facilitating the entry of air into injured gastric or perigastric vessels. Additionally, needle puncture and guidewire manipulation may create transient communication between the gastrointestinal lumen and the vascular system. These factors may have contributed to the development of systemic air embolism in this case.

Hyperbaric oxygen therapy is considered the mainstay of treatment for cerebral air embolism. A recent meta-analysis demonstrated that earlier initiation of therapy is significantly associated with improved neurological outcomes, highlighting the importance of prompt treatment.⁴ Clinically, sudden cardiopulmonary collapse during PEG is often attributed to aspiration. However, when chest radiography is unremarkable and neurologic recovery is delayed, air embolism should be considered. Early recognition is critical because timely hyperbaric oxygen therapy can significantly affect prognosis.

In conclusion, we report a rare case of cerebral air embolism occurring during PEG insertion using the pull method. Despite the absence of an identifiable intracardiac shunt, systemic embolization occurred, likely due to transpulmonary passage

of air. Clinicians should be aware of this rare but life-threatening complication, and prompt diagnosis with early hyperbaric oxygen therapy is essential.

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Ethics Statement

This case report was conducted in accordance with the principles of the Declaration of Helsinki. Written informed consent for publication of clinical information and images was obtained from the patient's legal guardian.

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